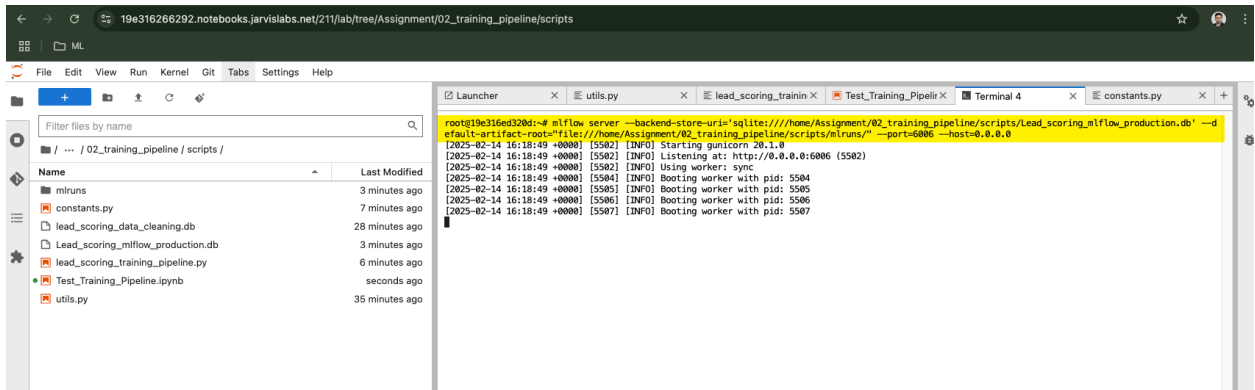


## ML OPS Training Pipeline

### Running the ML Flow Server



The screenshot shows a JupyterLab environment with a file browser on the left and a terminal on the right. The file browser displays the contents of the `02_training_pipeline/scripts` directory, including files like `mlruns`, `constants.py`, `lead_scoring_data_cleaning.db`, `lead_scoring_mlflow_production.db`, `lead_scoring_training_pipeline.py`, `Test_Training_Pipeline.ipynb`, and `utils.py`. The terminal window shows the command to start the ML Flow server and its output logs.

```
root@19e316ed320d:~# mlflow server --backend-store-uri=sqlite:///home/Assignment/02_training_pipeline/scripts/Lead_scoring_mlflow_production.db' --d
efault-artifact-root="file:///home/Assignment/02_training_pipeline/scripts/mlruns/" --port=6006 --host=0.0.0.0
[2025-02-14 16:18:49 +0000] [5502] [INFO] Starting gunicorn 20.1.0
[2025-02-14 16:18:49 +0000] [5502] [INFO] Listening at: http://0.0.0.0:6006 (5502)
[2025-02-14 16:18:49 +0000] [5502] [INFO] Using worker: sync
[2025-02-14 16:18:49 +0000] [5504] [INFO] Booting worker with pid: 5504
[2025-02-14 16:18:49 +0000] [5505] [INFO] Booting worker with pid: 5505
[2025-02-14 16:18:49 +0000] [5506] [INFO] Booting worker with pid: 5506
[2025-02-14 16:18:49 +0000] [5507] [INFO] Booting worker with pid: 5507
```

### Experiments - Lead\_scoring\_mlflow\_production

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### Experiments

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- Lead\_scoring\_mlflow...

### Lead\_scoring\_mlflow\_production

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Track machine learning training runs in experiments. [Learn more](#)

Experiment ID: 2

Description [Edit](#)

[Refresh](#) [Compare](#) [Delete](#) [Download CSV](#) [Start Time](#) [All time](#)

[Columns](#) Only show differences ☐  [Search](#) [Filter](#) [Clear](#)

Showing 1 matching run

|                          | Start Time     | Duration | Run Name    | User | Source      | Version | Models     | False Negative | Precision | Precision_0 | boosting_type | class_weight |
|--------------------------|----------------|----------|-------------|------|-------------|---------|------------|----------------|-----------|-------------|---------------|--------------|
| <input type="checkbox"/> | 21 seconds ago | 8.8s     | run_LightGB | root | ipykerne... | -       | LightGBM/4 | 6190           | 0.739     | 0.789       | gbdt          | None         |

[Load more](#)

19e3162662921.notebooks.jarvislabs.net/#/experiments/2/runs/93fe3ca12aa648519d5c5e735a91a479

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### Lead\_scoring\_mlflow\_production > run\_LightGB

run\_LightGB

Date: 2025-02-14 22:01:06 Source: ipykernel\_launcher.py User: root

Duration: 8.8s Status: FINISHED Lifecycle Stage: active

Description [Edit](#)

Parameters (20)

Metrics (12)

Tags

Artifacts

models

- MLmodel
- conda.yaml
- model.pkl
- python\_env.yaml
- requirements.txt

Full Path: file:///home/Assignment/02\_training\_pipeline/scripts/mlruns/2/93fe3ca12aa648519d5c5e735a91a479/artifacts/models

LightGBM, v4 Registered on 2025/02/14

### MLflow Model

The code snippets below demonstrate how to make predictions using the logged model. This model is also registered to the [model registry](#).

#### Model schema

Input and output schema for your model. [Learn more](#)

| Name                                                                                                   | Type |
|--------------------------------------------------------------------------------------------------------|------|
| No schema. See <a href="#">MLflow docs</a> for how to include input and output schema with your model. |      |

#### Make Predictions

Predict on a Spark DataFrame:

```
import mlflow
logged_model = 'runs:/93fe3ca12aa648519d5c5e735a91a479/models'

# Load model as a Spark UDF. Override result_type if the model does not return double values.
loaded_model = mlflow.pyfunc.spark_udf(spark, model_uri=logged_model, result_type='double')

# Predict on a Spark DataFrame.
columns = list(df.columns)
df.withColumn('predictions', loaded_model(*columns)).collect()
```

19e3162662921.notebooks.jarvislabs.net/#/models/LightGBM

mlflow 1.26.1 Experiments Models GitHub Docs

Registered Models > LightGBM

### LightGBM

Created Time: 2025-02-14 21:49:18 Last Modified: 2025-02-14 22:17:18

► Description [Edit](#)

► Tags

▼ Versions All Active 1 Compare

| <input type="checkbox"/> | Version                   | Registered at       | Created by | Stage      | Description |
|--------------------------|---------------------------|---------------------|------------|------------|-------------|
| <input type="checkbox"/> | <a href="#">Version 4</a> | 2025-02-14 22:01:13 |            | Production |             |
| <input type="checkbox"/> | <a href="#">Version 3</a> | 2025-02-14 21:59:14 |            | None       |             |
| <input type="checkbox"/> | <a href="#">Version 2</a> | 2025-02-14 21:49:42 |            | None       |             |
| <input type="checkbox"/> | <a href="#">Version 1</a> | 2025-02-14 21:49:18 |            | None       |             |

< 1 >

19e3162662921.notebooks.jarvislabs.net/#/experiments/2/runs/93fe3ca12aa648519d5c5e735a91a479

mlflow 1.26.1 Experiments Models GitHub Docs

Lead\_scoring\_mlflow\_production > run\_LightGB

### run\_LightGB

Date: 2025-02-14 22:01:06 Source: `ipykernel_launcher.py` User: root

Duration: 8.8s Status: FINISHED Lifecycle Stage: active

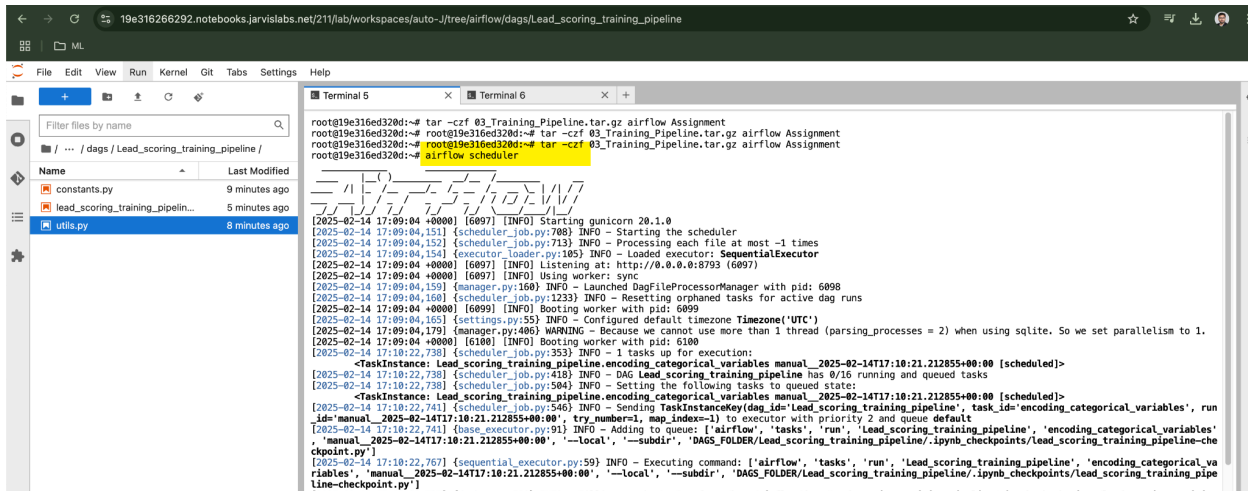
► Description [Edit](#)

► Parameters (20)

▼ Metrics (12)

| Name                             | Value |
|----------------------------------|-------|
| False Negative <a href="#">🔗</a> | 6190  |
| Precision <a href="#">🔗</a>      | 0.739 |
| Precision_0 <a href="#">🔗</a>    | 0.789 |
| Precision_1 <a href="#">🔗</a>    | 0.706 |
| Recall <a href="#">🔗</a>         | 0.747 |
| Recall_0 <a href="#">🔗</a>       | 0.649 |
| Recall_1 <a href="#">🔗</a>       | 0.829 |
| True Negative <a href="#">🔗</a>  | 23095 |
| auc <a href="#">🔗</a>            | 0.747 |
| f1_0 <a href="#">🔗</a>           | 0.712 |
| f1_1 <a href="#">🔗</a>           | 0.762 |

# Airflow Server Running for Training Pipeline

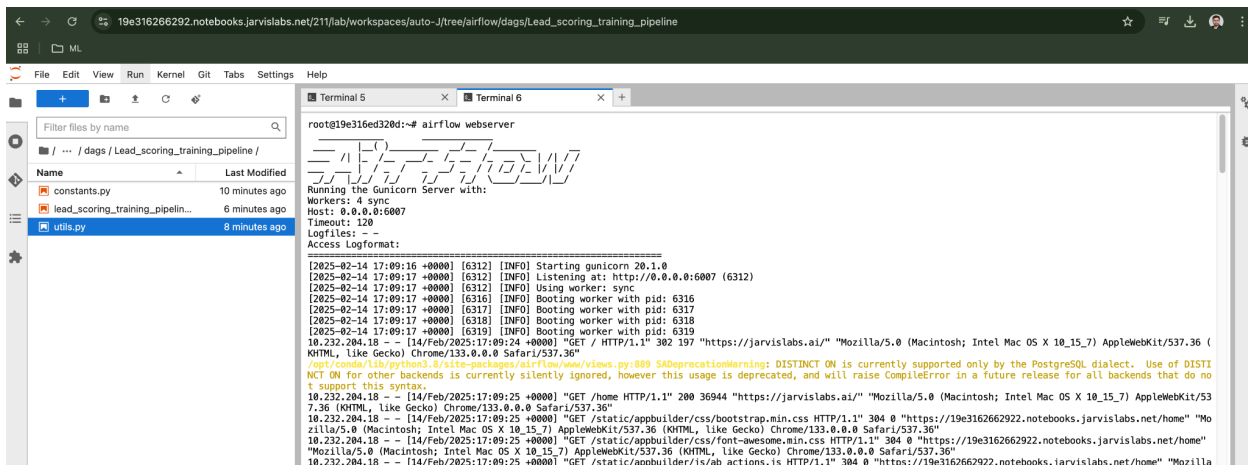


```

root@19e316ed320d:~# tar -czf 03_Training_Pipeline.tar.gz airflow Assignment
root@19e316ed320d:~# root@19e316ed320d:~# tar -czf 03_Training_Pipeline.tar.gz airflow Assignment
root@19e316ed320d:~# root@19e316ed320d:~# tar -czf 03_Training_Pipeline.tar.gz airflow Assignment
root@19e316ed320d:~# airflow scheduler

[2025-02-14 17:09:04 +0000] [6097] [INFO] Starting unicorn 20.1.0
[2025-02-14 17:09:04.151] [scheduler_job.py:708] INFO - Starting the scheduler
[2025-02-14 17:09:04.152] [scheduler_job.py:713] INFO - Processing each file at most -1 times
[2025-02-14 17:09:04.154] [executor_loader.py:105] INFO - Loaded executor: SequentialExecutor
[2025-02-14 17:09:04 +0000] [6097] [INFO] Listening at: http://0.0.0.0:8793 (6097)
[2025-02-14 17:09:04 +0000] [6097] [INFO] Using worker: sync
[2025-02-14 17:09:04.150] [manager.py:160] INFO - Launched DagFileProcessorManager with pid: 6098
[2025-02-14 17:09:04.160] [scheduler_job.py:1233] INFO - Resetting orphaned tasks for active dag runs
[2025-02-14 17:09:04 +0000] [6099] [INFO] Booting worker with pid: 6099
[2025-02-14 17:09:04.165] [settings.py:55] INFO - Configured default timezone Timezone('UTC')
[2025-02-14 17:09:04.179] [manager.py:406] WARNING - Because we cannot use more than 1 thread (parsing_processes = 2) when using sqlite. So we set parallelism to 1.
[2025-02-14 17:09:04 +0000] [6100] [INFO] Booting worker with pid: 6100
[2025-02-14 17:10:22.738] [scheduler_job.py:353] INFO - 1 tasks up for execution:
-taskInstance: Lead_scoring_training_pipeline.encoding_categorical_variables manual_2025-02-14T17:10:21.212855+00:00 [scheduled]>
[2025-02-14 17:10:22.738] [scheduler_job.py:418] INFO - DAG Lead_scoring_training_pipeline has 0/16 running and queued tasks
[2025-02-14 17:10:22.738] [scheduler_job.py:504] INFO - Setting the following tasks to queued state:
-taskInstance: Lead_scoring_training_pipeline.encoding_categorical_variables manual_2025-02-14T17:10:21.212855+00:00 [scheduled]>
[2025-02-14 17:10:22.741] [scheduler_job.py:546] INFO - Sending TaskInstanceKey(dag_id='Lead_scoring_training_pipeline', task_id='encoding_categorical_variables', run_id='manual_2025-02-14T17:10:21.212855+00:00', try_number=1, map_index=-1) to executor with priority 2 and queue default
[2025-02-14 17:10:22.741] [base_executor.py:91] INFO - Adding to queue: ['airflow', 'tasks', 'run', 'Lead_scoring_training_pipeline', 'encoding_categorical_variables', 'manual_2025-02-14T17:10:21.212855+00:00', '--local', '--subdir', 'DAGS_FOLDER/Lead_scoring_training_pipeline/.ipynb_checkpoints/Lead_scoring_training_pipeline-checkpoint.py']
[2025-02-14 17:10:22.767] [sequential_executor.py:59] INFO - Executing command: ['airflow', 'tasks', 'run', 'Lead_scoring_training_pipeline', 'encoding_categorical_variables', 'manual_2025-02-14T17:10:21.212855+00:00', '--local', '--subdir', 'DAGS_FOLDER/Lead_scoring_training_pipeline/.ipynb_checkpoints/Lead_scoring_training_pipeline-checkpoint.py']

```



```

root@19e316ed320d:~# airflow webserver

Running the Unicorn Server with:
Host: 0.0.0.0:8087
Timeout: 120
Logfiles: -
Access LogFormat:

[2025-02-14 17:09:16 +0000] [6312] [INFO] Starting unicorn 20.1.0
[2025-02-14 17:09:17 +0000] [6312] [INFO] Listening at: http://0.0.0.0:8087 (6312)
[2025-02-14 17:09:17 +0000] [6312] [INFO] Using worker: sync
[2025-02-14 17:09:17 +0000] [6316] [INFO] Booting worker with pid: 6316
[2025-02-14 17:09:17 +0000] [6317] [INFO] Booting worker with pid: 6317
[2025-02-14 17:09:17 +0000] [6318] [INFO] Booting worker with pid: 6318
[2025-02-14 17:09:17 +0000] [6319] [INFO] Booting worker with pid: 6319
10.232.204.18 - - [14/Feb/2025:17:09:24 +0000] "GET / HTTP/1.1" 302 197 "https://jarvislabs.ai/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/133.0.0.0 Safari/537.36"
10.232.204.18 - - [14/Feb/2025:17:09:25 +0000] "GET /static/appbuilder/css/bootstrap.min.css HTTP/1.1" 304 0 "https://19e3162662922.notebooks.jarvislabs.net/home" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/133.0.0.0 Safari/537.36"
10.232.204.18 - - [14/Feb/2025:17:09:25 +0000] "GET /static/appbuilder/css/font-awesome.min.css HTTP/1.1" 304 0 "https://19e3162662922.notebooks.jarvislabs.net/home" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/133.0.0.0 Safari/537.36"
10.232.204.18 - - [14/Feb/2025:17:09:25 +0000] "GET /static/appbuilder/js/ab_actions.js HTTP/1.1" 304 0 "https://19e3162662922.notebooks.jarvislabs.net/home" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/133.0.0.0 Safari/537.36"

```

19e3162662922.notebooks.jarvislabs.net/home

Airflow DAGs Security Browse Admin Docs 17:11 UTC TK

Do not use SQLite as metadata DB in production – it should only be used for dev/testing. We recommend using Postgres or MySQL. [Click here](#) for more information.

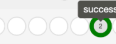
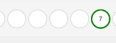
Do not use SequentialExecutor in production. [Click here](#) for more information.

## DAGs

All 34 Active 2 Paused 32

Filter DAGs by tag

Search DAGs

| DAG                                                                                                                         | Owner   | Runs                                                                              | Schedule | Last Run             | Next Run             | Recent Tasks                                                                       | Actions                                                                                                                                                                 | Links |
|-----------------------------------------------------------------------------------------------------------------------------|---------|-----------------------------------------------------------------------------------|----------|----------------------|----------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|  Lead_scoring_training_pipeline            | airflow |  | @monthly | 2025-02-14, 17:10:21 | 2025-03-01, 00:00:00 |  |   | ...   |
|  TK_Lead_Scoring_Data_Engineering_Pipeline | airflow |  | @daily   | 2025-02-14, 14:31:34 | 2025-02-14, 00:00:00 |  |   | ...   |

Showing 1-2 of 2 DAGs

19e3162662922.notebooks.jarvislabs.net/dags/Lead\_scoring\_training\_pipeline/grid

Airflow DAGs Security Browse Admin Docs 17:10 UTC TK

### DAG: Lead\_scoring\_training\_pipeline

Training pipeline for Lead Scoring System

Schedule: @monthly Next Run: 2025-03-01, 00:00:00

Grid Graph Calendar Task Duration Task Tries Landing Times Details <> Code Audit Log

14/02/2025, 05:10:39 PM 25 All Run Types All Run States Clear Filters

Auto-refresh

Duration: 00:00:23

encoding\_categorical\_variables  
training\_model

DAG Lead\_scoring\_training\_pipeline

DAG Details

DAG Runs Summary

|                      |                          |
|----------------------|--------------------------|
| Total Runs Displayed | 1                        |
| Total running        | 1                        |
| First Run Start      | 2025-02-14, 17:10:22 UTC |
| Last Run Start       | 2025-02-14, 17:10:22 UTC |
| Max Run Duration     | 00:00:23                 |
| Mean Run Duration    | 00:00:23                 |
| Min Run Duration     | 00:00:23                 |

DAG Summary

|                 |   |
|-----------------|---|
| Total Tasks     | 2 |
| PythonOperators | 2 |

Version: v2.3.3

19e3162662922.notebooks.jarvislabs.net/dags/Lead\_scoring\_training\_pipeline/graph

Airflow DAGs Security Browse Admin Docs 17:11 UTC TK

**DAG: Lead\_scoring\_training\_pipeline** Training pipeline for Lead Scoring System success Schedule: @monthly Next Run: 2025-03-01, 00:00:00

Grid Graph Calendar Task Duration Task Tries Landing Times Gantt Details <> Code Audit Log

2025-02-14T17:10:22Z Runs 25 Run manual\_\_2025-02-14T17:10:21.212855+00:00 Layout Left > Right Update Find Task...

PythonOperator

deferred failed queued running scheduled skipped success up\_for\_reschedule up\_for\_retry upstream\_failed no\_status

Auto-refresh

```
graph LR; encoding_categorical_variables --> training_model;
```